# Process Transfer Plan

**A-Mab Drug Substance — PTP-001**

Process Development / Manufacturing Science & Technology (MSAT)

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**SYNTHETIC DOCUMENT — NOT A REAL RECORD.** Illustrative content generated from a seeded model of the *A-Mab* case study (CMC Biotech Working Group, 2009). Novacyte Biologics and all sites, SOP/AMV numbers and signatories are fictional. For NLP corpus development only; not for any regulated use.

## Approvals

Table 1: Approval record (synthetic; signatures not applied).

| Role | Function | Name (synthetic) | Signature | Date |
|---|---|---|---|---|
| Author | Process Development / MSAT | — | — | — |
| Reviewer | Analytical Development | — | — | — |
| Reviewer | Manufacturing Science | — | — | — |
| Reviewer | Statistics | — | — | — |
| Approver | Quality Assurance | — | — | — |

## Abbreviations

ADCC, antibody-dependent cellular cytotoxicity; AEX, anion exchange chromatography; AMV, analytical method validation; BLA, Biologics License Application; CEX, cation exchange chromatography; CPP, critical process parameter; CQA, critical quality attribute; DNA, deoxyribonucleic acid; DO, dissolved oxygen; DoE, design of experiments; DS, drug substance; ELISA, enzyme-linked immunosorbent assay; GPP, general process parameter; HCP, host cell protein; HILIC, hydrophilic interaction liquid chromatography; icIEF, imaged capillary isoelectric focusing; IgG1, immunoglobulin G subclass 1; KPP, key process parameter; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; PAR, proven acceptable range; pCO2, dissolved carbon dioxide partial pressure; PD, Process Development; PPQ, process performance qualification; QA, Quality Assurance; QC, Quality Control; qPCR, quantitative polymerase chain reaction; RSM, response-surface methodology; SEC, size-exclusion chromatography; SOP, standard operating procedure; TFF, tangential flow filtration; UF/DF, ultrafiltration and diafiltration; WC-CPP, well-controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

# 1 Purpose and scope

This plan governs the transfer of the A-Mab drug substance manufacturing process from Novacyte Biologics — Cambridge, MA (Development) to Novacyte Biologics — Grafton, WI (Commercial DS). A-Mab is a humanized IgG1 monoclonal antibody (anti-CD20-like) whose principal mechanism of action is antibody-dependent cellular cytotoxicity (ADCC), so the glycan attributes that govern effector function are formed in the production bioreactor and are not modified by the purification train of the drug substance process (CMC Biotech Working Group 2009). The process was developed at the sending site and is now to be established as the licensed commercial process at the receiving site, at a production bioreactor working volume of 15,000 L. The transfer covers the 8 drug substance unit operations (Steps 3 to 10), from the production bioreactor through the final ultrafiltration and diafiltration step, together with the in-process controls, the analytical methods and the control strategy that govern them.

Technology transfer under ICH Q10 is the transfer of product and process knowledge between development and manufacturing, so that the receiving site can realise the product (International Council for Harmonisation 2008). The knowledge that has to move is more than a process description. It is the understanding of which parameters affect which quality attributes, over what ranges and with what assurance, and under the lifecycle approach to process validation that understanding is the Stage 1 output (process design) that becomes the input to the Stage 2 qualification of the commercial process (U.S. Food and Drug Administration 2011).

This plan is the parent document of the A-Mab process characterization package. Every characterization plan and report in that package exists to supply part of the knowledge the transfer requires, namely the operating ranges the receiving site will run to (the normal operating ranges, or NORs), the classification of each process parameter, and the control strategy that assures the critical quality attributes (CQAs) at commercial scale. The dependency runs in one direction. The risk assessment RA-001 derives from this plan and sets the scope of the studies, the master plan PCMP-001 governs their execution, and the master report PCMR-001 consolidates their outcome into the package that supports the Biologics License Application (BLA). The documents are listed in Table 1.1.

Table 1.1: The A-Mab document package parented by this plan.

| Document ID | Document class | Subject |
|---|---|---|
| RA-001 | Pre-Characterization Process Risk Assessment | A-Mab Drug Substance |
| PCMP-001 | Process Characterization Master Plan | A-Mab Drug Substance |

| Document ID | Document class | Subject |
|---|---|---|
| PCP-003 | Process Characterization Plan | Production Bioreactor (Step 3) |
| PCR-003 | Process Characterization Report | Production Bioreactor (Step 3) |
| PCP-004 | Process Characterization Plan | Harvest and Clarification (Step 4) |
| PCR-004 | Process Characterization Report | Harvest and Clarification (Step 4) |
| PCP-005 | Process Characterization Plan | Protein A Chromatography (Step 5) |
| PCR-005 | Process Characterization Report | Protein A Chromatography (Step 5) |
| PCP-006 | Process Characterization Plan | Low-pH Viral Inactivation (Step 6) |
| PCR-006 | Process Characterization Report | Low-pH Viral Inactivation (Step 6) |
| PCP-007 | Process Characterization Plan | Cation Exchange Chromatography (Step 7) |
| PCR-007 | Process Characterization Report | Cation Exchange Chromatography (Step 7) |
| PCP-008 | Process Characterization Plan | Anion Exchange Chromatography (Step 8) |
| PCR-008 | Process Characterization Report | Anion Exchange Chromatography (Step 8) |
| PCP-009 | Process Characterization Plan | Small-Virus Retentive Filtration (Step 9) |
| PCR-009 | Process Characterization Report | Small-Virus Retentive Filtration (Step 9) |
| PCP-010 | Process Characterization Plan | Ultrafiltration / Diafiltration (Step 10) |
| PCR-010 | Process Characterization Report | Ultrafiltration / Diafiltration (Step 10) |
| PCMR-001 | Process Characterization Master Report | A-Mab Drug Substance |

Three areas are outside the scope of this plan. Drug product manufacture, its control strategy and its own transfer are covered separately and are not addressed here. The execution and reporting of the process performance qualification (PPQ) campaign is also excluded, since this plan defines only the prerequisites for that campaign and the strategy behind it, while the PPQ protocol and the PPQ report are issued as separate controlled documents. Finally, the plan is prospective. It records no characterization result, and it makes no claim about the outcome of any study named in Table 1.1.

# 2 Product and process description

A-Mab is produced in fed-batch mammalian cell culture and purified on the platform train that Novacyte uses for its humanized IgG1 products (CMC Biotech Working Group 2009). The drug substance is delivered at 75 g/L after ultrafiltration and diafiltration (Step 10), and development experience at the sending site gives a nominal harvest titre of about 4.4 g/L, which the receiving site is expected to reproduce when it operates the same fed-batch process at 15,000 L working volume.

The purification train is Protein A capture, low-pH viral inactivation, cation exchange (CEX), anion exchange (AEX), small-virus retentive filtration, and ultrafiltration and diafiltration (UF/DF). This ordering is fixed and is used in the same form in every document of the package. The steps and their principal roles in the control strategy are given in Table 2.1.

Table 2.1: The A-Mab drug substance process train and the principal role of each step.

| Step | Unit operation | Principal role |
|---:|---|---|
| 3 | Production Bioreactor | Forms the glycan, charge-variant and aggregate CQAs (design-space step) |
| 4 | Harvest and Clarification | Primary recovery and clarification; forms no product-quality CQA |
| 5 | Protein A Chromatography | Capture; sets leached Protein A; principal HCP and DNA clearance |
| 6 | Low-pH Viral Inactivation | Low-pH hold; sets the cumulative XMuLV (enveloped) clearance |
| 7 | Cation Exchange Chromatography | Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance |
| 8 | Anion Exchange Chromatography | Flow-through polish; sets the cumulative MVM clearance; clears XMuLV/HCP/DNA |
| 9 | Small-Virus Retentive Filtration | Dedicated small-virus removal; principal MVM clearance |
| 10 | Ultrafiltration / Diafiltration | Formulation / mass balance; delivers drug substance at target concentration |

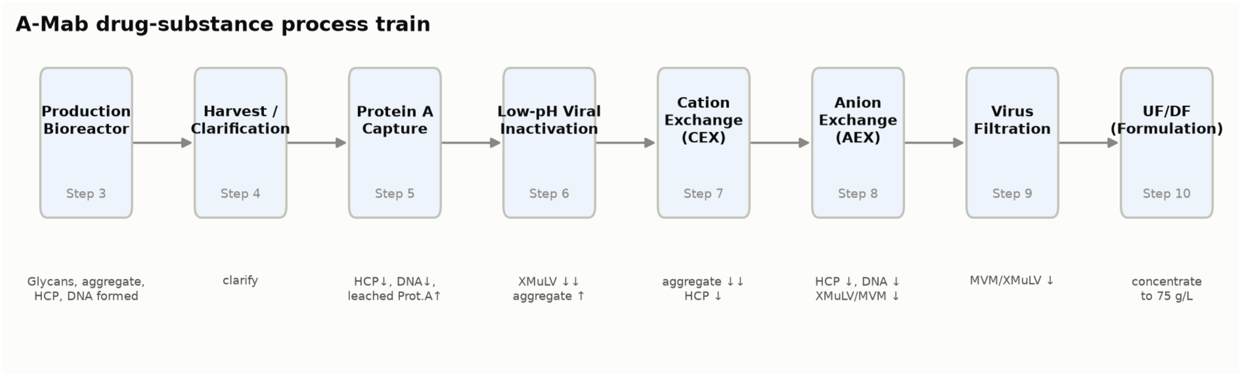


Figure 2.1: The A-Mab drug substance process train (Steps 3 to 10), with the principal quality impact of each unit operation.

Two features of the train shape the characterization effort that supports this transfer. The production bioreactor (Step 3) forms the glycan, charge variant and aggregate attributes of the drug substance (afucosylation, galactosylation, high mannose, acidic charge variants and aggregates), so it carries the largest study of the campaign and it defines most of the design space. Viral safety, by contrast, is modular. The low-pH hold (Step 6), the anion exchange step (Step 8) and the virus filtration step (Step 9) each contribute an independent increment of clearance, and no single step carries the cumulative claim on its own (International Council for Harmonisation 2023a). Two steps form no product quality attribute at all: harvest and clarification (Step 4) is a primary recovery operation, and ultrafiltration and diafiltration (Step 10) is a concentration and buffer exchange operation that delivers the drug substance at its target concentration.

The drug substance is defined by 10 quality attributes, of which 6 are of high or very high criticality and are therefore treated as critical (CMC Biotech Working Group 2009). Criticality is assigned on a continuum (very low to very high) and not as a binary label, because the uncertainty around a structure-function relationship is itself part of the assessment (International Council for Harmonisation 2023b). The attributes, their acceptance criteria and their criticality are given in Table 2.2. High mannose and the two cumulative viral clearance attributes (XMuLV for enveloped virus and MVM for small non-enveloped virus) carry the very high assignment, and the acceptance criteria given for them are the limits the transferred process must meet at commercial scale. Every characterization study in the package is assessed against those criteria.

Table 2.2: A-Mab drug substance quality attributes, acceptance criteria and criticality. Tool #1 is the impact-by-uncertainty criticality score of the case study (CMC Biotech Working Group 2009).

| CQA | Category | Acceptance | Criticality | Tool #1 |
|---|---|---|---|---|
| Afucosylation | glycosylation | 2–13 % of glycans | H | 60 |
| Galactosylation (total %Gal) | glycosylation | 10–40 % (G1+G2 equiv.) | H | 48 |

| CQA | Category | Acceptance | Criticality | Tool #1 |
|---|---|---|---|---|
| High Mannose | glycosylation | 3–10 % of glycans | VH | 80 |
| Aggregates (HMW) | size variant | 0–5 % HMW (SEC) | H | 60 |
| Acidic charge variants (deamidation) | charge variant | 18–40 % (CEX/icIEF) | VL | 4 |
| Host Cell Protein (HCP) | process impurity | 0–100 ng/mg | M-H | 36 |
| Residual DNA | process impurity | 0–0.001 ng/dose | VL | 6 |
| Leached Protein A | process impurity | 0–5 ppm | L-M | 16 |
| Viral clearance — XMuLV (enveloped) | viral safety | 16.7–30 log10 (cumulative) | VH | 20 |
| Viral clearance — MVM (parvovirus) | viral safety | 8.6–20 log10 (cumulative) | VH | 20 |

# 3 Sending and receiving sites

Novacyte Biologics — Cambridge, MA (Development) is the sending site. It developed the process, holds the qualified scale-down models, and will execute the characterization studies listed in Table 1.1. Novacyte Biologics — Grafton, WI (Commercial DS) is the receiving site, and it will manufacture commercial drug substance for supply.

Process development, scale-down model qualification and design of experiments (DoE) work remain at the sending site for the duration of the transfer and are not relocated by this plan. Routine manufacture, in-process control and release testing move to the receiving site once the analytical method transfers and the qualification activities described in §6 are complete (see §6.2 for the analytical sequence and §7 for the qualification prerequisites). The division of activities between the two sites is given in Table 3.1.

Table 3.1: Division of activities between the sending and receiving sites.

| Activity | Sending site | Receiving site |
|---|---|---|
| Process development and design of experiments | Owns | Reviews |
| Scale-down model qualification | Owns | Reviews |
| Process characterization studies | Owns | Reviews |
| Master batch record and process description | Authors | Issues and executes |
| Analytical method validation | Owns | Receives by transfer |
| In-process and release testing | Interim | Routine after transfer |
| Engineering and PPQ batches | Supports | Owns |
| Change control and deviation management | Local quality system | Local quality system |

# 4 Site equivalency analysis

The receiving site can operate the A-Mab process within the ranges the sending site will characterize, subject to the differences set out below and to the qualification sequence in §7. The comparison covers the dimensions that determine whether a range established at one site holds at the other, which are the equipment train and its scale, the process control capability, the utilities and buffer supply, the analytical capability and the quality system. It is summarized in Table 4.1.

Table 4.1: Site equivalency comparison and how each difference is managed.

| Dimension | Sending site | Receiving site | Management of the difference |
|---|---|---|---|
| Equipment train | Bench and pilot scale-down train | Commercial train at 15,000 L | Scale-down qualification under SOP-1001 (§5) |
| Process control | Closed-loop control of pH, temperature, DO, feed | Closed-loop control on commercial instrumentation | Control capability against NOR shown in engineering runs |
| Utilities and buffers | Development-scale buffer preparation | Commercial buffer preparation under SOP-2103 | Buffer hold time and expiry verified at scale (§8) |
| Analytical capability | Validated methods, 10 AMV records | Methods received by transfer | Method transfer and requalification (§6.2) |
| Quality system | Novacyte pharmaceutical quality system | Novacyte pharmaceutical quality system | Common change control and deviation management |

Scale is the substantive difference and the reason the rest of this plan exists. The sending site operates bench and pilot equipment, and the receiving site operates the commercial train, so equivalency is claimed at the level of mechanism and control capability and not as identity of equipment. Scale-independent parameters (pH, temperature, dissolved oxygen, culture duration, buffer composition) are set to the same values at both sites, while scale-dependent parameters (agitation, gas flow, vessel pressure, $pCO_2$ accumulation, column bed height, load per unit membrane area) are matched on process performance and not on setting, following the scale-down qualification approach described in §5 (CMC Biotech Working Group 2009).

Process control capability is comparable in kind but has not been measured at the receiving site for this process. Both sites hold the culture parameters by closed-loop control, and both hold chromatography conductivity and pH by in-line measurement and feedback. The question that matters for the control strategy is narrower than that. It is whether the receiving site can hold each parameter inside its normal operating range with the margin that the classification of that parameter assumes, and that question will be answered by the engineering runs (§7) and not by this analysis.

Analytical capability differs today and is the second real gap. The 10 validated methods that support the process and its characterization are held at the sending site, and the receiving site laboratory has not yet executed any of them for A-Mab. The methods, together with the 19 procedures that govern the process and the assays, are listed in Table 6.1. The transfer path for each method is given in §6.2, and the residual position is carried in the gap analysis (§8).

The quality system is common to both sites, which removes a class of transfer risk that would otherwise have to be managed. Change control, deviation management, document control and training operate under one Novacyte pharmaceutical quality system, so a change made at either site is assessed against the same procedure. Movement within an approved design space is not a regulatory change, but it still requires a prospective assessment of the associated risks under the change management system, and that obligation applies at the receiving site from the day the process arrives (International Council for Harmonisation 2008; CMC Biotech Working Group 2009).

The equivalency argument above is bounded in two ways, and neither bound is closed by the analysis itself. It rests on qualified scale-down models and on platform experience with the earlier products (X-Mab, Y-Mab and Z-Mab), and not on commercial-scale A-Mab data, because no A-Mab batch has been produced at the receiving site. It also covers only the parameters carried in the process description, so a scale-dependent effect arising from a variable outside that set would not be detected by this comparison. Both bounds are closed by the engineering runs and the qualification sequence described in §7.

# 5 Scale-down model and comparability strategy

Characterization will be executed on scale-down models and not at commercial scale, so the qualification of those models carries the weight of the transfer. Qualification follows SOP-1001. A model is qualified when the scale-independent parameters are operated at the proposed commercial set-points, the scale-dependent parameters are set to match commercial performance, and the resulting process performance and quality attributes (step yield, impurity clearance and the attributes of Table 2.2) are statistically comparable across scales (CMC Biotech Working Group 2009). Where the adequacy of a model cannot be shown, the model and its predictions are modified before any conclusion is drawn from it (U.S. Food and Drug Administration 2011).

Each unit operation has its own scale-down model and its own qualification record. The qualification evidence for a step is reported in that step's characterization report (PCR-003 through PCR-010) and it is not repeated in this plan. The instrumented systems of the scale-down train that are held under individual calibration and change control are listed in Table 5.1, and there are 3 of them.

Table 5.1: Instrumented scale-down systems under calibration and change control at the sending site.

| Equipment | Description |
|---|---|
| EQ-CHR-118 | temperature-controlled chromatography chamber |
| EQ-BRX-205 | production scale-down bioreactor pCO2 probe |
| EQ-TFF-142 | ultrafiltration / diafiltration TFF skid |

The characterization scope that these models support is given in Table 5.2. Of the 37 process parameters in the register, 22 are assigned to multivariate study and 15 to univariate study, and that assignment is made in RA-001 on the basis of the potential impact of each parameter on the critical quality attributes and its potential to interact with other parameters (International Council for Harmonisation 2023b). Two rows of Table 5.2 carry no multivariate study. Harvest and clarification (Step 4) forms no product quality attribute, and ultrafiltration and diafiltration (Step 10) is a mass balance operation, so both steps are characterized univariately and neither will report a design space.

Table 5.2: Characterization scope by process step, with the plan and report that cover it.

| Step | Unit operation | Parameters | Multivariate | Univariate | Plan | Report |
|---|---|---|---|---|---|---|
| 3 | Production Bioreactor | 9 | 5 | 4 | PCP-003 | PCR-003 |
| 4 | Harvest and Clarification | 3 | 0 | 3 | PCP-004 | PCR-004 |
| 5 | Protein A Chromatography | 6 | 4 | 2 | PCP-005 | PCR-005 |
| 6 | Low-pH Viral Inactivation | 4 | 3 | 1 | PCP-006 | PCR-006 |
| 7 | Cation Exchange Chromatography | 5 | 4 | 1 | PCP-007 | PCR-007 |
| 8 | Anion Exchange Chromatography | 5 | 4 | 1 | PCP-008 | PCR-008 |
| 9 | Small-Virus Retentive Filtration | 2 | 2 | 0 | PCP-009 | PCR-009 |
| 10 | Ultrafiltration / Diafiltration | 3 | 0 | 3 | PCP-010 | PCR-010 |

Comparability between the two sites will be demonstrated in three stages, and the stages are cumulative. First, each scale-down model is qualified under SOP-1001 against sending-site pilot and commercial-representative data, which establishes that the characterization data describe the commercial process and not the model. Second, engineering runs at the receiving site are compared with the scale-down predictions at set-point, on the attributes of Table 2.2 and on step yield. Third, the PPQ batches confirm at commercial scale that the process holds its ranges and meets the acceptance criteria. The acceptance criteria for the second and third stages are defined in PCMP-001 and in the PPQ protocol respectively, and not in this plan.

Comparability at set-point does not confirm the edges of a design space. The design space will be defined from response-surface models fitted on scale-down data, and the engineering and PPQ batches will be run at target conditions, so the edges of the characterized region remain supported by the scale-down models alone (see §8). That limitation is restated as a gap below, and it constrains what the receiving site may claim when it later moves within the design space.

# 6 Transfer of the process, methods and control strategy

## 6.1 Manufacturing process

The manufacturing process transfers as a documented set, and the set is complete only when the receiving site can execute it without reference to the sending site. That set comprises the process description, the master batch record, the set-point and range tables for every unit operation, and the procedures that govern the operations and the assays. The 19 procedures and 10 method validations that make up the controlled-document basis of the transfer are listed in Table 6.1, and each is authored or re-issued under the receiving site's document control before the first engineering run. MSAT authors the process description, receiving-site Manufacturing reviews it for executability, and Quality Assurance (QA) approves it.

Table 6.1: Controlled procedures and validated analytical methods in the scope of the transfer. SOP and AMV numbers are synthetic placeholders.

| Reference | Title | Type |
|---|---|---|
| SOP-1001 | Qualification of Scale-Down Models | SOP |
| SOP-1002 | Design, Execution and Statistical Analysis of DoE Studies | SOP |
| SOP-2003 | Operation of the Production Bioreactor (Fed-Batch) | SOP |
| SOP-2004 | Cell-Culture Media and Feed Preparation | SOP |
| SOP-2005 | Harvest by Continuous Disk-Stack Centrifugation | SOP |
| SOP-2006 | Clarification by Depth and Sterile Filtration | SOP |
| SOP-2007 | Operation of the Protein A Capture Chromatography Step | SOP |
| SOP-2008 | Chromatography Resin Life-Cycle, Packing and Sanitization | SOP |
| SOP-2009 | Operation of the Low-pH Viral Inactivation Step | SOP |
| SOP-2010 | Operation of the Cation-Exchange Polishing Chromatography Step | SOP |

| Reference | Title | Type |
|---|---|---|
| SOP-2011 | Operation of the Anion-Exchange Polishing Chromatography Step | SOP |
| SOP-2012 | Operation and Post-Use Integrity Testing of the Small-Virus Retentive Filtration Step | SOP |
| SOP-2013 | Operation of the Ultrafiltration / Diafiltration (Formulation) Step | SOP |
| SOP-3010 | Released N-Glycan Mapping by 2-AB HILIC-UPLC | SOP |
| SOP-3011 | Size-Variant Analysis by SEC-HPLC | SOP |
| SOP-3012 | Host-Cell-Protein Quantitation by ELISA | SOP |
| SOP-3013 | Charge-Variant Analysis by icIEF | SOP |
| SOP-3014 | Residual Host-Cell DNA by qPCR | SOP |
| SOP-4001 | Process-Parameter Classification and Control-Strategy Definition | SOP |
| AMV-3010 | N-Glycan Map (2-AB HILIC-UPLC) | Method validation |
| AMV-3011 | Size-Variants (SEC-HPLC) | Method validation |
| AMV-3012 | Host-Cell Protein ELISA | Method validation |
| AMV-3013 | Charge Variants (icIEF) | Method validation |
| AMV-3014 | Residual DNA (qPCR) | Method validation |
| AMV-3015 | Turbidity by Nephelometry (NTU) | Method validation |
| AMV-3016 | Leached Protein A by ELISA (ppm) | Method validation |
| AMV-3017 | XMuLV Infectivity Titre (TCID50) | Method validation |
| AMV-3018 | MVM Infectivity Titre (TCID50/qPCR) | Method validation |
| AMV-3019 | Protein Concentration by A280 (UV) | Method validation |

The process description delivered at transfer carries set-points and proposed normal operating ranges for all 37 parameters. It does not carry proven acceptable ranges (PARs), and it does not carry parameter classifications, because both are outputs of the characterization studies and neither exists when this plan takes effect. The batch record issued for PPQ is the revised version that carries them, and that revision is the last step of the process transfer and not the first.

## 6.2 Analytical methods

Analytical method transfer is a separate activity from process transfer and it runs on its own schedule. Each of the 10 validated methods listed in Table 6.1 transfers under its own protocol, which specifies the samples, the number of analysts and the comparative criteria to be met against the validated performance recorded in the corresponding method validation (AMV) record. A method is transferred when the receiving site laboratory has met those criteria and Quality Assurance has approved the transfer report.

Until the transfer report for a method is approved, in-process and release testing that depends on that method remains at the sending site. This is why method transfer precedes the engineering runs in the sequence of §10. Method variance propagates directly into the capability estimates that justify the control strategy, so the comparative criteria are set against the validated precision and accuracy of each method (as recorded in the AMV) and not against a nominal target. Analytical Development at the sending site owns the transfer protocol, Quality Control (QC) at the receiving site executes it, and Quality Assurance approves the outcome.

## 6.3 Control strategy

The control strategy is the last element to transfer, because it is the only element that does not yet exist. Its content is fixed by the outcome of characterization, which is to say by which parameters are shown to affect a critical quality attribute, over which proven acceptable ranges, and with which residual assurance at commercial scale (International Council for Harmonisation 2009, 2012). Classification follows SOP-4001. A parameter whose variability affects a critical quality attribute is classified as a critical process parameter (CPP) or a well-controlled critical process parameter (WC-CPP) according to the risk that it falls outside the design space, and a parameter that affects process consistency without affecting quality is classified as a key process parameter (KPP) (CMC Biotech Working Group 2009).

Delivery follows the same order. Each characterization report classifies the parameters of its own step, PCMR-001 consolidates the classifications and the process design space, and the control strategy document and the revised master batch record follow PCMR-001. Quality Assurance approves the control strategy before PPQ. No PPQ batch is executed against an unclassified parameter set, and that constraint is what sequences the whole package.

# 7 Process performance qualification strategy

5 consecutive PPQ batches at commercial scale are planned at the receiving site. PPQ is Stage 2 of the process validation lifecycle, and it confirms at scale the process design established in Stage 1 (U.S. Food and Drug Administration 2011). It does not establish that design. The characterization package described in this plan is the Stage 1 record, so the strategy below states what that package must deliver before the first PPQ batch is charged.

The prerequisites for starting PPQ are the following.

- An approved characterization report exists for every unit operation (PCR-003 through PCR-010).
- PCMR-001 is approved, and it consolidates the design space, the parameter classifications and the projected commercial-scale capability against the acceptance criteria of Table 2.2.
- The control strategy and the revised master batch record are approved and issued under the receiving site's document control.
- Transfer reports are approved for all 10 analytical methods used for in-process control and release testing.
- Scale-down model qualification is complete and reported for every step under SOP-1001.
- Engineering runs at the receiving site have been executed and compared with the scale-down predictions at set-point.
- A readiness review has been held and its actions have been closed.

The PPQ protocol defines the sampling plan, the in-process and release testing and the acceptance criteria for each batch, and none of those is set in this plan. Capability against the acceptance criteria of Table 2.2 is projected in PCMR-001 from the characterization data and a commercial-scale variance model, and the PPQ batches test those projections against real manufacturing variability. A small number of consecutive batches cannot adequately capture the process variability expected at commercial manufacturing scale, so PPQ is treated as a confirmation of the design and not as the whole of the evidence for it (CMC Biotech Working Group 2009).

Stage 3 continued process verification begins on approval of the PPQ report and continues for the commercial life of the product (U.S. Food and Drug Administration 2011). Its design, including the trending of the parameters classified in PCMR-001, is outside the scope of this plan.

# 8 Gap analysis

Six gaps were identified between what the receiving site holds today and what it needs in order to manufacture A-Mab under an approved control strategy. Each is stated below with its impact on the transfer, the action that closes it, and the function that owns that action. None of the six is closed by this plan, and none is closed by the equivalency analysis of §4. They are summarized in Table 8.1.

No point in the proposed design space will have been run at commercial scale when the characterization package completes. The studies are executed on scale-down models at the sending site, the engineering and PPQ batches are run at set-point, and the edges of the characterized region are therefore supported by the scale-down models alone. The impact is that a scale-dependent effect at the edge of the region (for example pCO2 accumulation at commercial working volume, or a longer mixing time in a larger vessel) would not appear in the characterization data at all. The closing action is the qualification of every scale-down model under SOP-1001, with statistical comparison of performance across scales, followed by engineering runs and PPQ batches at set-point. MSAT at both sites owns the action. The residual position after closure is that operation at the edges of the design space at commercial scale requires a prospective risk assessment under the change management system before it is performed (International Council for Harmonisation 2008).

None of the 10 validated analytical methods has been executed for A-Mab at the receiving site. In-process and release testing cannot therefore be performed there when this plan takes effect, and the first engineering runs would otherwise generate data on instruments and by analysts that have never been compared against the sending site. The impact falls on the control strategy as well as on release, because method variance is part of the variance that the capability projections in PCMR-001 must account for. The closing action is the method transfer sequence of §6.2, with comparative precision and accuracy criteria for each method and an approved transfer report for each. Analytical Development at the sending site owns the protocol and Quality Control at the receiving site owns the execution. Until each report is approved, the affected testing stays at the sending site.

The control strategy that the receiving site must implement does not yet exist. All 37 parameters carry a proposed set-point and a proposed normal operating range, but none carries a final classification, and no proven acceptable range has been established. The impact is that the master batch record cannot be issued in its final form and PPQ cannot start, so the transfer cannot be completed on the process description alone. The closing action is the document sequence of Table 1.1 (RA-001, then the plans PCP-003 to PCP-010, then the reports PCR-003 to PCR-010, then PCMR-001), followed by classification under SOP-4001 and revision of the batch record. MSAT owns the sequence and Quality Assurance approves its output. This gap is the reason the PPQ prerequisites of §7 are written as a gate.

Viral clearance cannot be measured in the commercial facility. Spiking studies are performed only at scale-down and under containment, so the clearance claimed for the low-pH hold, for anion exchange and for virus filtration will rest entirely on scale-down data, with no confirmation at 15,000 L. The impact is that the cumulative XMuLV and MVM claims of Table 2.2 are supported by a modular argument across three steps and by the qualification of three scale-down models, and not by any commercial-scale measurement. The closing action has two parts. The scale-down models for Steps 6, 8 and 9 are qualified to worst-case commercial conditions, and the modular clearance studies are run to ICH Q5A (International Council for Harmonisation 2023a). The receiving site then demonstrates in the engineering runs that it can hold the governing parameters of those steps (hold pH and hold time, load pH and conductivity, and volumetric throughput) within the ranges the studies claim. MSAT owns the first part with virology support, and receiving-site Manufacturing owns the second. The residual position is that the claim remains modular. No single step is claimed to deliver it, and the consolidation across steps is reported in PCMR-001.

The Protein A capture step cannot inherit its ranges from platform data. RA-004 assessed the alternate Protein A resin and concluded that independent characterization is required, with no bridging from the platform data that supports the other steps of the train. The impact is that the operating ranges of the capture step, and the leached Protein A attribute that the step sets, have no prior-knowledge basis and must be established experimentally before the transfer can complete. The closing action is the full characterization of the step in PCP-005 and PCR-005, including the resin life-cycle and sanitization conditions governed by SOP-2008. Process Development at the sending site owns the action. Platform experience with the earlier products is retained as supporting context and is not used as a substitute for the step's own data.

The receiving site has no execution history for this process. No operator there has run an A-Mab batch, no buffer has been prepared at commercial scale to the A-Mab formulations, and the buffer hold times that the chromatography steps depend on have not been verified at that scale. The impact is concentrated in the first batches, where an execution error or an out-of-specification buffer lot would move the conductivity and pH of the chromatography loads, which are the governing parameters of the polishing and viral steps. The closing action is training against the transferred procedures of Table 6.1, buffer preparation and expiry verification under SOP-2103, at least one engineering run per unit operation, and a readiness review before PPQ. Receiving-site Manufacturing owns the action with MSAT support. Until the verification is complete, buffer expiry at the receiving site is held to the shortest period justified by the sending site data.

Table 8.1: Transfer gaps, closing actions, owners and the evidence that closes each.

| Gap | Closing action | Owner | Evidence of closure |
|---|---|---|---|
| Design space not confirmed at commercial scale | Scale-down qualification, engineering runs, PPQ at set-point | MSAT (both sites) | SOP-1001 qualification records; PCMR-001 |
| Analytical methods not qualified at the receiving site | Method transfer protocol and report per method | Analytical Development / QC | Approved transfer report per AMV |
| No parameter classification or control strategy | RA-001, PCP-003 to PCP-010, PCR-003 to PCR-010, PCMR-001 | MSAT / QA | Approved control strategy and batch record |
| Viral clearance measurable only at scale-down | Modular clearance studies to ICH Q5A; parameter control demonstrated | MSAT / Virology | PCR-006, PCR-008, PCR-009; PCMR-001 |
| Protein A resin has no platform bridging (RA-004) | Independent characterization of the capture step | Process Development | PCP-005 and PCR-005 |
| No execution history at the receiving site | Training, buffer verification under SOP-2103, engineering runs | Receiving-site Manufacturing | Readiness review record |

# 9 Roles and responsibilities

The sending site owns the process knowledge and the characterization package. The receiving site owns commercial manufacture and the execution of the engineering and PPQ batches. Quality Assurance at both sites approves the transferred documents and the release of the resulting material. The responsibilities are given in Table 9.1.

Table 9.1: Roles and responsibilities for the transfer.

| Function | Site | Responsibility |
|---|---|---|
| Process Development | Sending | Process knowledge, scale-down models, characterization studies |
| MSAT | Sending and receiving | Transfer strategy, scale-down qualification, gap closure |
| Analytical Development | Sending | Method validation and method transfer protocols |
| Quality Control | Receiving | Method execution, in-process and release testing |
| Manufacturing | Receiving | Batch record execution, engineering and PPQ batches |
| Quality Assurance | Sending and receiving | Document approval, change control, batch disposition |
| Statistics | Sending | Design of experiments, capability projection, comparability analysis |

# 10 Deliverables and schedule

This plan commits to the deliverables listed in Table 10.1. The sequence is fixed by dependency and not by calendar date, because each deliverable is a prerequisite for the one below it. No dates are given here. The transfer schedule is maintained in the project plan and is revised under change control.

Table 10.1: Deliverables committed by this plan and the dependency each one satisfies.

| Deliverable | Owner | Prerequisite for |
|---|---|---|
| RA-001 pre-characterization risk assessment | MSAT | Scope of every characterization plan |
| PCMP-001 characterization master plan | MSAT | Execution of PCP-003 to PCP-010 |
| PCP-003 to PCP-010 characterization plans | Process Development | Execution of the studies |
| Scale-down model qualification records | Process Development | Validity of the study data |
| PCR-003 to PCR-010 characterization reports | Process Development | PCMR-001 |
| Analytical method transfer reports | Analytical Development | Testing at the receiving site |
| PCMR-001 characterization master report | MSAT | Control strategy and PPQ |
| Control strategy and revised master batch record | MSAT / QA | Engineering and PPQ batches |
| Engineering run report | Receiving-site Manufacturing | PPQ readiness review |

# 11 Approvals

This plan is approved by the functions recorded in Table 1, and any change to the transfer strategy, the gap list or the PPQ prerequisites requires revision of the plan and re-approval by the same functions. The scope of each approval is given in Table 11.1.

Table 11.1: Scope of approval by function.

| Function | Scope of approval |
|---|---|
| Process Development / MSAT | Transfer strategy, equivalency analysis, gap list |
| Analytical Development | Analytical method transfer approach |
| Manufacturing Science | Executability at the receiving site |
| Statistics | Comparability and capability approach |
| Quality Assurance | Compliance with the pharmaceutical quality system |

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